单条件

454 加载使用

访存2 执行2	访存1 执行1	访存0 执行0
W	M	E
×	×	D
×	D	
D		

D_stall F_stall

d_srcA == W_dstE

6种

方法 F_stall , D_stall, E_bubble

条件 (d_srcA in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
d_srcB in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM}) &&
E_icode in {IRMOVQ, IMRMOVQ, IRRMOVQ, IRMMOVQ, IOPQ, IPUSHQ}

错误分支预测

1种

×
×
E ← 错误的跳转
×
×

条件 E_icode == JXX && !e_Cnd

方法 D_bubble、E_bubble

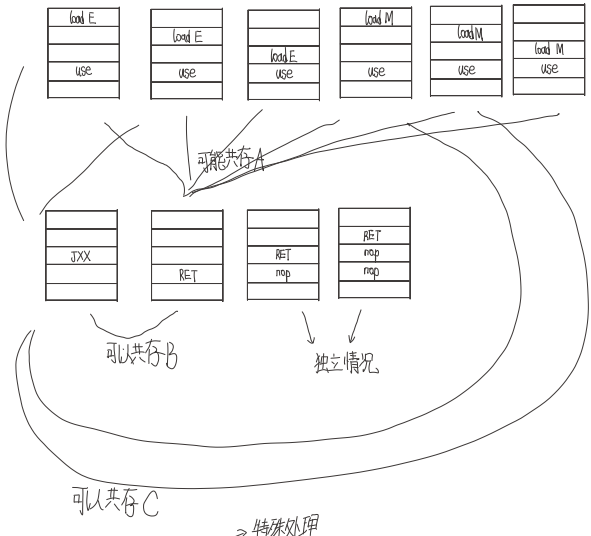
RET

×	×	×
×	×	M
×	E	×
D	×	×
×	×	×

3种

条件: RET in {D_icode、E_icode、M_icode}

方法: F_stall, D_bubble



特殊处理

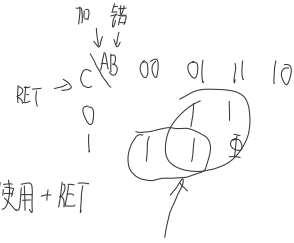
组合A: F_stall、D_stall、E_bubble

组合B: F_stall, D_bubble、E_bubble

组合C: F_stall、D_bubble、E_bubble

F_bubble = 0

F_stall = d_srcA in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
d_srcB in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
RET in {D_icode, E_icode, M_icode}



D_stall = (d_srcA in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
d_srcB in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM}) &&
!(JXX && !e_Cnd)

→单加载使用、加载使用 + RET

D_bubble = (JXX && !e_Cnd) || RET in {D_icode, E_icode, M_icode}

→单错误预、单RET、错误预测 + RET，错误预 + 加载使用

E_bubble = d_srcA in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
d_srcB in {E_dstE, E_dstM, M_dstE, M_dstM, W_dstE, W_dstM} ||
(JXX && !e_Cnd)

%rdi、%rsi
↓
%rdx、%rcx

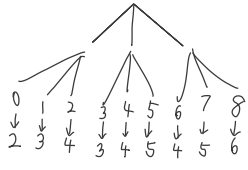
xor %rdx, %rcx
↓
0 1 0

11 → 0
01 → 1
10 → 1
00 → 2

and
xor

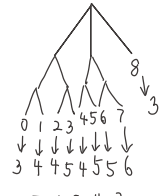
线性 0 1 2 3 4 5 6 7 8
↓
平均4次

树状



(9 + 12 + 15)/9 = 4

load use优化 9



7+9+9+11+3
(36+3)/9